Altitude-Matched Retrieval and Agentic Mixture-of-Experts for a Local-First Medical Reasoning Companion

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***Abstract*—Retrieval-augmented generation (RAG) has become the dominant paradigm for grounding large language models (LLMs) in domain knowledge, yet two structural gaps persist for consumer medical question answering. First, flat retrieval returns leaf passages that answer specific questions well but cannot serve *broad* questions whose answer lives only in the aggregate of many passages. Second, single-embedding dense retrieval is brittle to the colloquial, under-specified phrasing typical of patients. We present Healix, a fully on-device medical reasoning companion, and its retrieval stack, which addresses both gaps with a single mechanism we call *altitude-matched retrieval*: one small language model drafts a *hypothetical* answer at two levels of abstraction, and each level is routed to the matching tier of a hierarchical index—a specific hypothetical searches the leaf passage index, while a broad hypothetical searches a RAPTOR-style summary tree. This couples Hypothetical Document Embeddings (HyDE) and recursive summarization in a way that, to our knowledge, has not been described. We further introduce an *agentic Mixture-of-Medical-Experts* (MoME) router that performs system-level sparse gating over specialist retrieval lenses. The complete system runs offline on a single commodity GPU. On a query-variation retrieval benchmark derived from MedQuAD and PubMedQA, we report the effect of each stage on Recall@k and MRR, and we quantify the latency engineering—most notably a $3\times$ reduction in query-side overhead—required to make the stack interactive. We discuss the safety posture of a disclaimer-free medical assistant and release the full system as open source.**

***Index Terms*—retrieval-augmented generation, dense retrieval, hierarchical retrieval, mixture of experts, medical question answering, on-device inference, large language models**

I. Introduction

Large language models can produce fluent medical explanations but hallucinate facts they were not reliably trained on, and privacy concerns discourage sending personal health questions to hosted APIs. Retrieval-augmented generation (RAG) [1] mitigates the first problem by grounding generation in retrieved evidence; running the entire pipeline locally mitigates the second. Building such a system for lay medical use, however, exposes two failure modes that standard RAG does not handle.

The altitude gap. Users ask questions at very different levels of abstraction. "What is the maximum daily dose of acetaminophen?" is answered by a single leaf passage. "How does chronic stress affect the body overall?" is not: no single passage contains the answer, which is an aggregate over cardiovascular, immune, digestive, and neurological sources. Flat dense retrieval, tuned to return the top passages, structurally cannot serve the second class. Recursive summarization indexes such as RAPTOR [3] add higher-level summary nodes to close this gap on the *index* side, but still search those nodes with the raw user query, whose embedding sits far from any summary.

The phrasing gap. Patients phrase questions colloquially and incompletely ("my chest feels tight when I'm stressed"), whereas the corpus is written in clinical register. Single-vector dense retrieval blurs exact terms and is sensitive to this mismatch. Hypothetical Document Embeddings (HyDE) [2] address this on the *query* side by embedding a generated hypothetical answer instead of the question, but classic HyDE produces one hypothetical against one flat index.

Our key observation is that these two mechanisms are complementary along the same axis—abstraction—and can be fused. We generate a hypothetical at *two* altitudes in a single language-model call and route each to the index tier whose granularity it matches. The specific hypothetical is concatenated with the query for leaf retrieval; the broad hypothetical searches the summary tree. This *altitude matching* aligns query and corpus geometry on both axes simultaneously, and the marginal cost of the second altitude is zero because both are produced by the same call.

We integrate altitude-matched retrieval into Healix, a local-first medical companion, alongside a system-level sparse Mixture-of-Medical-Experts router, a hybrid dense/sparse retriever with reranking, and a persona-controlled generator, all served offline from a single GPU. Our contributions are:

1) **Altitude-matched retrieval**, a novel coupling of HyDE and RAPTOR in which multi-granularity hypotheticals are routed to matching index tiers (Section IV).
2) **Agentic Mixture-of-Medical-Experts (MoME)**, an embedding-gated sparse router over specialist retrieval lenses that expands evidence coverage without a second generation pass (Section V).
3) **A fully local, disclaimer-free deployment** with latency engineering that makes the multi-stage stack interactive on commodity hardware, and a query-variation evaluation on MedQuAD/PubMedQA quantifying each stage (Sections VI–VII).

II. Related Work

Dense and hybrid retrieval. Dense passage retrieval [4] maps queries and passages to a shared vector space; we use BGE embeddings [8] over a FAISS [9] index. Lexical BM25 [5] remains competitive for exact terms (drug names, rare conditions); fusing the two with Reciprocal Rank Fusion [6] and cross-encoder reranking [7] is now standard practice, and Anthropic's Contextual Retrieval [10] shows that prepending situating context before indexing further improves recall. Healix uses all of these as its base retriever.

Query- and index-side augmentation. HyDE [2] embeds a generated hypothetical answer to bridge the query–corpus register gap. RAPTOR [3] builds a tree of recursive summaries so that broad queries can retrieve aggregated evidence. Both are single-altitude: HyDE produces one hypothetical; RAPTOR searches its tree with the raw query. We are not aware of prior work that generates hypotheticals at multiple abstraction levels and matches each to a corresponding index tier.

Mixture of experts and agents. Sparsely-gated MoE [11] and Switch Transformers [12] route tokens to expert subnetworks *inside* a model. More recently, mixture-of-agents approaches [13] route at the system level across LLM instances. Healix applies sparse gating one level up again—over medical *specialties*, each with its own retrieval lens—so that routing shapes evidence gathering rather than parameter selection.

Medical QA and safety. MedQuAD [14] and PubMedQA [15] are widely used medical QA corpora and form our knowledge base and evaluation source. Large medical LLMs such as Med-PaLM [16] target clinician-grade benchmarks; Healix instead targets lay users on-device, trading model scale for privacy, controllability, and grounded retrieval.

III. System Architecture

Healix processes each turn through the pipeline in Fig. 1 (described textually here). An input message first passes a *scope gate* that resolves greetings, identity, and clearly non-medical requests to canned responses with zero model cost, admitting only medical queries to the full pipeline. For an admitted query the system (i) runs the MoME router to select specialist lenses, (ii) generates two-altitude HyDE hypotheticals, (iii) retrieves leaf evidence via hybrid dense/sparse search under each expert lens and overview evidence from the RAPTOR tree, (iv) composes a single grounded prompt, and (v) streams a persona-controlled answer from a local LLM. Persistent conversation memory supplies continuity across turns. All components are served offline; the LLM runs on a local Ollama server and every index is memory-mapped from disk.

```
message → scope gate → [medical]
  → MoME router (specialist lenses)
  → two-altitude HyDE (one small-LM call)
  → leaf: hybrid dense+BM25+RRF+rerank
  → broad: RAPTOR summary-tree search
  → prompt compose → local LLM stream
```

Fig. 1. The Healix per-turn pipeline. Altitude-matched retrieval (Section IV) spans the two retrieval lines; the MoME router (Section V) conditions the leaf line.

IV. Altitude-Matched Retrieval

A. Hierarchical index

The leaf tier L_0 is the corpus of N=239,644 passage chunks embedded with BGE and indexed in FAISS using 8-bit scalar quantization, chosen because it preserves absolute cosine scores (mean score error 1.7×10^{-4} versus a flat index) so that score thresholds remain valid. We construct two summary tiers offline. Leaf embeddings are reconstructed directly from the quantized index (no re-embedding) and clustered with k-means into K_1=1500 topic clusters; the local LLM summarizes each cluster's representative members into a titled overview paragraph, forming tier L_1. The L_1 summaries are themselves embedded and clustered into K_2=120 domain nodes (L_2). The build is resumable and processes clusters largest-first so partial builds cover the highest-mass topics. The resulting tree contains $1500+114=1614$ overview nodes.

B. Two-altitude hypotheticals

Given query q, a single small-LM call emits two hypotheticals: a *specific* sentence h_s carrying concrete clinical detail, and a *broad* sentence h_b naming the overarching topic. We deliberately use a sub-billion-parameter model here because the hypothetical is consumed only as an embedding target, not shown to the user; a lenient parser falls back to the raw generation when a small model ignores the output format, so the mechanism never silently no-ops.

C. Matched routing

Leaf retrieval embeds the concatenation $[q; h_s]$ and searches L_0 through the full hybrid pipeline. Overview retrieval embeds h_b (falling back to q) and searches $L_1 \cup L_2$; the top overview nodes are *prepended* to the evidence block so the generator reads the frame before the details. Formally, for tier L_t with granularity g_t, we search with the hypothetical h whose target granularity is closest to g_t:

$$\text{evidence} = \bigcup_t \text{TopK}\big(L_t,\ \phi(h_{\arg\min_a |g_a - g_t|})\big),$$

where ϕ is the shared embedding function. Classic HyDE is the special case of a single tier and single hypothetical; classic RAPTOR is the special case of searching all tiers with $\phi(q)$. Matching the hypothetical's altitude to the tier aligns query and corpus geometry on both axes at once, and because both hypotheticals come from one call, the second altitude is free. The mechanism is *fail-open*: any error, a missing tree, or an unavailable LM degrades gracefully to standard hybrid retrieval on q.

V. Agentic Mixture-of-Medical-Experts

Rather than route tokens (as in intra-model MoE) or whole LLM instances (as in mixture-of-agents), Healix routes over medical *specialties*. Each expert e carries (i) a semantic prototype p_e encoded once, (ii) a keyword prior, and (iii) a retrieval *lens*—a query reformulation template. The gate

TABLE I
END-TO-END QUERY LATENCY, WARM, SINGLE RTX-4060-CLASS GPU (MEAN OVER HELD-OUT MEDICAL QUERIES).

Configuration	Retrieval (s)	Total (s)	Time-to-first-token (s)
HyDE on 7B model	8–10	≈22	≈18
HyDE on 0.5B model	3–4	≈10	≈6

computes cosine similarity between the query embedding and each prototype, adds keyword and imaging-modality boosts, and applies a sharpened softmax; the top-k experts above a floor are activated:

$$w_e = \text{softmax}_\tau\big(\langle \phi(q), p_e \rangle + \beta \mathbb{1}[\text{kw}] + \gamma \mathbb{1}[\text{modality}]\big).$$

Each activated expert reformulates q through its lens and gathers specialty-specific evidence; the union grounds a *single* synthesis pass framed as an integrated panel, so coverage improves without the cost of multiple generations. The current roster covers Cardiology, Pulmonology, Neurology, Dermatology, Gastroenterology, Musculoskeletal, Pharmacology, Psychophysiology, and a default General expert. Activated experts and gate weights are surfaced to the user as an explainability signal.

VI. IMPLEMENTATION AND LATENCY ENGINEERING

Healix is served by a FastAPI backend streaming tokens over Server-Sent Events to a browser UI; generation uses a local Ollama server (default `qwen2.5:7b-instruct`). The knowledge index (175 MB quantized) and chunk store are memory-mapped. Because the multi-stage pipeline places several operations *before* the first generated token, query-side latency dominates perceived responsiveness. Profiling the live server isolated the two-altitude HyDE call as the bottleneck: invoking the full 7B model to draft throwaway embedding text cost ≈10 s, while MoME evidence gathering cost only ≈0.6 s. Because the hypothetical is used only for embedding, we moved HyDE to a sub-billion-parameter model with a lenient parser, cutting the call to ≈3 s. Table I reports the resulting end-to-end improvement; all components remain fail-open so any stage can be disabled or can fail without breaking a response.

VII. EVALUATION

A. Protocol

Because the corpus questions appear verbatim in the index, evaluating retrieval with the raw question is circular (dense retrieval trivially self-matches). We therefore use a *query-variation* protocol: we sample n question–chunk pairs, paraphrase each question into colloquial patient phrasing with an LLM, keep the *original chunk* as the gold target, and measure whether each retrieval configuration recovers it in the top k. This isolates robustness to phrasing, exactly the property hybrid and HyDE claim to improve. We report Recall@1/5/10 and MRR@10 for three configurations: **dense** (FAISS + cross-encoder rerank), **hybrid** (dense + contextual BM25 + RRF + rerank), and **hybrid+HyDE** (hybrid search over $[q; h_s]$). Sample size n, seed, and paraphrase model are fixed and released with the code.

TABLE II
QUERY-VARIATION RETRIEVAL ON PARAPHRASED MEDQUAD/PUBMEDQA (n=150, k=10, SEED 13, 100% PARAPHRASED). HIGHER IS BETTER; LAST COLUMN IS MEAN QUERY-SIDE LATENCY.

Configuration	R@1	R@5	R@10	MRR@10	Lat. (s)
Dense	**0.700**	**0.853**	0.880	**0.760**	0.14
Hybrid	**0.700**	0.820	0.880	0.755	0.19
Hybrid + HyDE	0.667	0.847	0.880	0.739	3.24

B. Retrieval results

Table II reports the query-variation benchmark, and the result is a candid one: on this single-target task the three configurations are statistically indistinguishable (all within ≈2 points on every metric; identical Recall@10 of 0.880), and adding HyDE does not improve recall while costing ≈3 s per query. We report this rather than a favorable subset. Three factors plausibly explain the parity. *(i)* The cross-encoder reranker is strong and largely equalizes the candidate sets that dense and hybrid produce, a known effect when reranking dominates first-stage recall. *(ii)* An LLM paraphrase of an in-domain question preserves most medical salience, so the dense encoder rarely fails badly enough for lexical or hypothetical augmentation to rescue it. *(iii)* The metric scores recovery of a *single* gold chunk; it does not measure the aggregate-evidence regime that the hierarchical arm targets, so it cannot credit altitude matching. The practical implication is concrete: on in-domain single-target retrieval, HyDE's latency is not justified by recall, which is precisely why we make it cheap (small model) and fail-open rather than mandatory. Demonstrating a retrieval win for hybrid/HyDE requires harder out-of-distribution or term-heavy queries (e.g., rare drug or gene names), which this in-domain benchmark does not contain; we flag this as the primary evaluation gap.

C. Hierarchical retrieval

The altitude-matched broad arm is evaluated qualitatively, as no single gold chunk exists for aggregate questions. On broad queries the summary tree returns on-topic overview nodes with high similarity (e.g., "how does stress affect the body" → *Stress and Brain Function*, *Cortisol and Stress Responses*; "managing diabetes" → *Preventing and Managing Diabetes*), whereas a 24-node ablation of the tree returned only loosely related nodes—indicating that coverage, not the mechanism, governs broad-arm quality. A gold-labeled aggregate-answer benchmark is left to future work.

D. System behavior

Under a 5-way concurrent load the server completed all requests with isolated conversation state and no errors, serializing generation behind a single lock. Adversarial and out-of-domain inputs (prompt-injection attempts, non-medical requests, empty and malformed messages) were refused or handled without leaking system instructions.

VIII. Discussion and Limitations

Safety posture. Healix is deliberately disclaimer-free and avoids ritual "consult a professional" closings, favoring specific, calm guidance. This design choice trades a conventional safety net for readability and must be revisited for any real deployment, particularly the handling of red-flag emergency presentations, where suppressing urgent guidance would be harmful. **Evaluation scope.** Our benchmark measures retrieval, not answer quality; a human or LLM-judge study of grounded answer accuracy is the most important next step. **Model ceiling.** The 7B generator bounds clinical nuance; the architecture is model-agnostic and benefits directly from stronger local models. **Paraphrase confound.** The query-variation protocol paraphrases questions with an LLM. The retriever's encoder is a separate embedding model, so this is less circular than an LLM-as-judge study, but paraphrase artifacts remain; a human-paraphrased query set would be stronger. We fix the seed and release all artifacts for reproduction.

IX. Conclusion

We presented Healix and *altitude-matched retrieval*, which couples HyDE and RAPTOR by generating multi-granularity hypotheticals and routing each to the matching tier of a hierarchical index, together with a system-level Mixture-of-Medical-Experts router. The complete stack runs offline on commodity hardware, and our latency engineering makes it interactive. We hope the altitude-matching idea—aligning query and corpus abstraction jointly—proves useful beyond the medical setting. Code and artifacts are released to support reproduction.

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